

Pipelining

- A machine learning pipeline can be created by putting together a sequence of steps involved in training a machine learning model.
- It can be used to automate a machine learning workflow.
- The pipeline can involve pre-processing, feature selection, classification/regression, and post-processing.
- More complex applications may need to fit in other necessary steps within this pipeline.
- Definition of pipeline class according to scikit-learn is Sequentially apply a list of transforms and a final estimator.
- Intermediate steps of pipeline must implement fit and transform methods and the final estimator only needs to implement fit.

```
class sklearn.pipeline.Pipeline(steps)
```

Pipeline of transforms with a final estimator.

Sequentially apply a list of transforms and a final estimator. Intermediate steps of the pipeline must be 'transforms', that is, they must implement fit and transform methods. The final estimator only needs to implement fit. The transformers in the pipeline can be cached using `memory` argument.

A machine learning pipeline is a series of interconnected data processing and modeling steps designed to automate, standardize and streamline the process of building, training, evaluating and deploying machine learning models.

A machine learning pipeline is a crucial component in the development and productionization of [machine learning](#) systems, helping [data scientists](#) and data engineers manage the complexity of the end-to-end machine learning process and helping them to develop accurate and scalable solutions for a wide range of applications.

Machine learning technology is advancing at a rapid pace, but we can identify some broad steps involved in the process of building and deploying machine learning and deep learning models.

1. **Data collection:** In this initial stage, new data is collected from various data sources, such as databases, APIs or files. This data ingestion often involves raw data which may require preprocessing to be useful.
2. **Data preprocessing:** This stage involves cleaning, transforming and preparing input data for modeling. Common preprocessing steps include handling missing values, encoding categorical variables, scaling numerical features and splitting the data into training and testing sets.
3. **Feature engineering:** Feature engineering is the process of creating new features or selecting relevant features from the data that can improve the model's predictive power. This step often requires domain knowledge and creativity.
4. **Model selection:** In this stage, you choose the appropriate machine learning algorithm(s) based on the problem type (e.g., classification, regression), data characteristics, and performance requirements. You may also consider hyperparameter tuning.

5. **Model training:** The selected model(s) are trained on the training dataset using the chosen algorithm(s). This involves learning the underlying patterns and relationships within the training data. Pre-trained models can also be used, rather than training a new model.
6. **Model evaluation:** After training, the model's performance is assessed using a separate testing dataset or through cross-validation. Common evaluation metrics depend on the specific problem but may include accuracy, precision, recall, F1-score, mean squared error or others.
7. **Model deployment:** Once a satisfactory model is developed and evaluated, it can be deployed to a production environment where it can make predictions on new, unseen data. Deployment may involve creating APIs and integrating with other systems.
8. **Monitoring and maintenance:** After deployment, it's important to continuously monitor the model's performance and retrain it as needed to adapt to changing data patterns. This step ensures that the model remains accurate and reliable in a real-world setting.

Machine learning lifecycles can vary in complexity and may involve additional steps depending on the use case, such as hyperparameter optimization, cross-validation, and feature selection. The goal of a machine learning pipeline is to automate and standardize these processes, making it easier to develop and maintain ML models for various applications.